



ರೈ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
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MNIST - Modified National Institute of Standards and technology database.

Report :- MNIST Handwritten Digit Classification Using TensorFlow / keras.

Introduction :-

Deep learning is a branch of Machine learning that uses artificial neural networks to learn patterns from data. In this assignment, the MNIST handwritten digit dataset is used to build a simple neural network using TensorFlow / keras. The purpose of the project is to classify handwritten digit images into one of ten classes from 0 to 9.

The Dataset contains grayscale images of handwritten digits, where each image has a size of 28×28 pixels. The dataset is divided into training and testing data. The training data is used to train the neural network while the testing data is used to evaluate its performance.

Dataset Exploration :-

The MNIST dataset was loaded using the built-in dataset available in TensorFlow / keras.

The dataset contains of :

- 60,000 training Images
- 10,000 testing Images
- Image Size : 28×28 pixels
- 10 classes: 0 to 9
- Image type : Grayscale

Each Image contains pixels values ranging from 0 to 255. Before training, the pixel values were normalized to the range 0 to 1 by dividing them by 255.

Model Design:-

A simple neural network was designed using Tensor Flow / keras.

The model contains the following layers:

Flatten layer :-

The Input Image has a size of 28×28 pixels. The Flatten layer converts the two-dimensional Image into a one-dimensional vector.

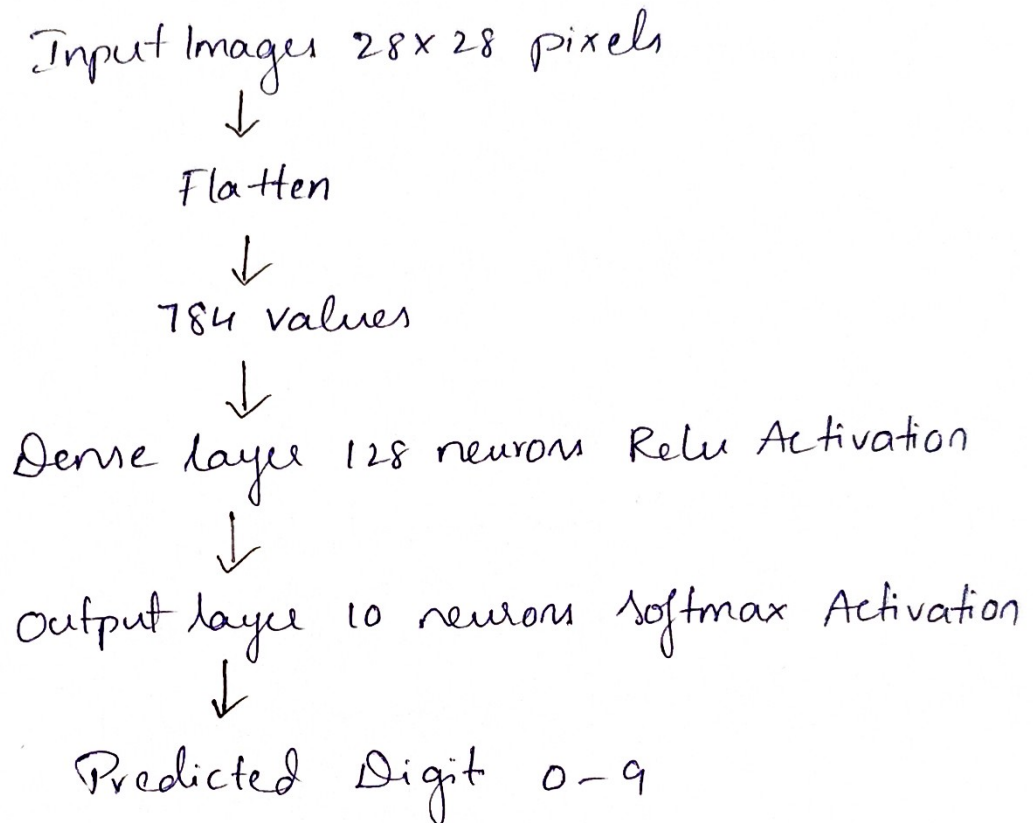
$$\therefore 28 \times 28 = 784 \text{ input values.}$$

Dense Hidden layer:- A fully connected Dense layer containing 128 neurons are used.

ReLU activation function was used in this layer.

output layer:- It contains 10 neurons, corresponding to the ten possible digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The softmax activation function is used to calculate the probability of each class.

Architecture:-



The model was trained for 10 epochs with a batch size of 32. A validation split of 20% was used during training.

Evaluation:-

After training, the model was evaluated using the 10,000 MNIST test images.

The evaluation was performed using Test loss & Test accuracy.

Test Accuracy: 97.610 % (0.9761)

Test loss : 0.0889

The test accuracy represents the percentage of test images that were correctly classified by neural network.

Training Visualization:

During model training, the training accuracy & validation accuracy were recorded for each epoch.

The accuracy values were used to visualize the learning performance of the model.

→ Training & validating Accuracy:- Accuracy generally increases as the model learns from the training data.

- Validation Accuracy shows how the model performs on data that was not directly used for updating the model weights.

→ Training & validation loss:- loss represents the error b/w the actual labels and predictions made by the model. The loss values are monitored during training to understand the learning behaviour of the neural network.

Testing on Sample Images:-

After training, the model was tested on 5 sample images from the MNIST test dataset.

The model predicted the digit for each image, and the predicted label were compared with actual labels.

Sample	Actual	Predicted	Correct?
1	7	7	✓
2	2	2	✓
3	1	1	✓
4	0	0	✓
5	4	4	✓

All 5 out of 5 images were correctly classified, demonstrating the model's practical reliability at recognising handwritten digits it had not seen during training.

Experiment Analysis:-

Experiment was performed by changing the number of neurons in the hidden layer.

Original model contains - 128 neurons

Dense (128, activation = 'relu')

Experimental model - 256 neurons

Increased to Dense (256, activation = 'relu')

The other parameters were kept unchanged, optimizer, loss function, Epochs, Batch size, validation split.

→ It shows, the effect of changing the number of neurons in the hidden layer. Increasing number of neurons changes the model's capacity to learn features and patterns from the handwritten images.

Conclusion: - The MNIST handwritten digit classification model was successfully implemented using TensorFlow and Keras. The model achieved a test accuracy of approx 98% (97.61%) and correctly predicted sample handwritten digits.

~~The~~ It helps to understand the effect of model architecture on accuracy. Overall, this assignment provided practical knowledge of ~~data~~ deep learning workflow from data preprocessing to prediction and evaluation.